Cao Thang Nguyen

Department of Mechanical Engineering
Ulsan National Institute of Science & Technology (UNIST)

Email: thangckt@gmail.com
Phone: +82 10 4993 5329
Website: thangckt.github.io
Profiles: GoogleScholar, ResearchGate, ORCID

EDUCATION

Ph.D. 2/2023	Mechanical Engineering, Ulsan National Institute of Science and Technology (UNIST), South Korea.
M.S. 8/2016	Mechanical Engineering, University of Ulsan, South Korea.
B.S. 5/2010	Mechanical Engineering, HCMC University of Technology and Education, Vietnam.

POSITIONS

2/2023 - present	Postdoctoral Researcher, Ulsan National Institute of Science & Technology (UNIST), South Korea.
3/2017 - 2/2023	Graduate Researcher, Ulsan National Institute of Science & Technology (UNIST), South Korea.
9/2014 - 8/2016	Graduate Researcher, University of Ulsan, Korea.
1/2010 - 9/2014	R&D Staff, Industrial Zones, Vietnam.

RESEARCH AREAS

- ➢ Computational mechanics and materials science.
- ➢ Structural/crystal elastic instabilities, phase transformation.
- ➢ Atomistic simulations (Molecular Dynamics, Enhanced Sampling, Density Functional Theory)

RESEARCH & TECHNICAL SKILLS

Computational Modeling & Simulation

- ✓ **Molecular Dynamics** (MD) and **Molecular Statics** (MS) simulations, including transition phenomena. [1–14]
- ✓ Free energy calculations using **Enhanced Sampling** methods (Steered MD, MetaDynamics, Mean-Force Dynamics) [5–7, 14] and **Thermodynamic Integration (TI)** [4, 5].
- ✓ **Density Functional Theory** (DFT) calculations for materials modeling [13, 15].
- ✓ Computational modeling of **2D materials** [6–8], **crystalline** [3–5, 9–11, 14], and **polymeric** [6] molecular systems.
- ✓ Application **Machine Learning** in computational materials science. [1, 2, 7, 13, 15, 16, 22, 23]
- ✓ Strong background in Finite Element Method (FEM), Solid Mechanics, Statistical Mechanics, Thermodynamics.

High-Performance Computing & Automation

- ✓ Compiling, configuring, and running large-scale open-source simulation packages on high-performance computing (HPC) clusters (LAMMPS, PLUMED, GPAW, OpenMPI, etc.) (example).
- ✓ Development of computational **pre- and post-processing pipelines** [15, 16, 22, 23, 17–21, 24–28]
- ✓ Design of **automated simulation workflows** across heterogeneous HPC resources [15, 23, 21, 25].
- ✓ Experience with multiple schedulers (SLURM, SGE)

Programming & Software

- ✓ Python (Advanced): scientific computing, data analysis, automation
- ✓ Modern Fortran (Basic): numerical kernels, legacy code interaction
- ✓ C++ (Basic), MATLAB (Intermediate), Bash
- ✓ Computational codes: LAMMPS, PLUMED, GPAW, ASE, DFTD3, OVITO, ABACUS
- ✓ Scientific Python libraries: NumPy, SciPy, Pandas, Polars, Matplotlib
- ✓ Development & reproducibility tools: Git/GitHub, Conda, Pip, Jupyter, LaTeX, BibTeX
- ✓ OS: CentOS, Rocky Linux, Fedora, Ubuntu, Windows, WSL

Certificates

2022 Research and Proposal Writing in the Sciences - AuthorAID

2013 SolidWorks Professional in Mechanical Design - Dassault Systems Corp

2012 Hard skills in Project Management - FMIT Institute

Notes

I am authorized to work in the US without need for visa sponsorship.

PUBLICATIONS

Journal Articles

[1] Fengbo Yuan, Zhaohan Ding, Yun-Pei Liu, Kai Cao, Jiahao Fan, Cao Thang Nguyen, Yuzhi Zhang, Haidi Wang, Yixiao Chen, Jiameng Huang, Tongqi Wen, Mingkang Liu, Yifan Li, Yong-Bin Zhuang, Hao Yu, Ping Tuo, Yaotang Zhang, Yibo Wang, Linfeng Zhang, Han Wang, and Jinzhe Zeng. "DPDispatcher: Scalable HPC Task Scheduling for AI-Driven Science". In: *Journal of Chemical Information and Modeling* (Nov. 2025), acs.jcim.5c02081. ISSN: 1549-9596, 1549-960X. DOI: 10.1021/acs.jcim.5c02081.

[2] Jinzhe Zeng, Xingliang Peng, Yong-Bin Zhuang, Haidi Wang, Fengbo Yuan, Duo Zhang, Renxi Liu, Yingze Wang, Ping Tuo, Yuzhi Zhang, Yixiao Chen, Yifan Li, Cao Thang Nguyen, Jiameng Huang, Anyang Peng, Marián Rynik, Wei-Hong Xu, Zezhong Zhang, Xu-Yuan Zhou, Tao Chen, Jiahao Fan, Wanrun Jiang, Bowen Li, Denan Li, Haoxi Li, Wenshuo Liang, Ruihao Liao, Liping Liu, Chenxing Luo, Logan Ward, Kaiwei Wan, Junjie Wang, Pan Xiang, Chengqian Zhang, Jinchao Zhang, Rui Zhou, Jia-Xin Zhu, Linfeng Zhang, and Han Wang. "Dpdata: A Scalable Python Toolkit for Atomistic Machine Learning Data Sets". In: *Journal of Chemical Information and Modeling* (Oct. 2025). ISSN: 1549-9596. DOI: 10.1021/acs.jcim.5c01767.

[3] Cao Thang Nguyen, Duc Tam Ho, and Sung Youb Kim. "Coalescence-enhanced melting in the incipient stage of surface melting". In: *Computational Materials Science* 242 (June 2024), p. 113092. DOI: 10.1016/j.commatsci.2024.113092.

[4] Cao Thang Nguyen, Duc Tam Ho, Viet Hung Ho, and Sung Youb Kim. “Calculation of melting temperature using nonequilibrium thermodynamic integration methods”. In: *Advanced Theory and Simulations* (2023), p. 2300588. DOI: 10.1002/adts.202300588.

[5] Cao Thang Nguyen, Duc Tam Ho, and Sung Youb Kim. “An Enhanced Sampling Approach for Computing the Free Energy of Solid Surface and Solid-Liquid Interface”. In: *Advanced Theory and Simulations* (2023), p. 2300538. DOI: 10.1002/adts.202300538.

[6] Van Huy Nguyen, Minwook Kim, Cao Thang Nguyen, Muhammad Suleman, Dinh Cong Nguyen, Naila Nasir, Malik Abdul Rehman, Hyun Min Park, Sohee Lee, Sung Youb Kim, Sunil Kumar, and Yongho Seo. “Fast fabrication technique for high-quality van der Waals heterostructures using inert shielding gas environment”. In: *Applied Surface Science* 639 (2023), p. 158186. ISSN: 0169-4332. DOI: 10.1016/j.apsusc.2023.158186.

[7] **Viet Hung Ho**, **Cao Thang Nguyen**, **Hoang D. Nguyen**, Hyun Suk Oh, Myoungsu Shin, and Sung Youb Kim. “Hydrogenated Graphene with Tunable Poisson’s Ratio Using Machine Learning: Implication for Wearable Devices and Strain Sensors”. In: *ACS Applied Nano Materials* 5.8 (2022), pp. 10617–10627. DOI: 10.1021/acsanm.2c01950.

[8] Viet Hung Ho, Duc Tam Ho, Cao Thang Nguyen, and Sung Youb Kim. “Negative out-of-plane Poisson’s ratio of bilayer graphane”. In: *Nanotechnology* 33.25 (2022), p. 255705. DOI: 10.1088/1361-6528/ac5da0.

[9] **Cao Thang Nguyen**, **Duc Tam Ho**, Seung Tae Choi, Doo-Man Chun, and Sung Youb Kim. “Pattern transformation induced by elastic instability of metallic porous structures”. In: *Computational Materials Science* 157 (2019), pp. 17–24. ISSN: 0927-0256. DOI: 10.1016/j.commatsci.2018.10.023.

[10] **Duc Tam Ho**, **Cao Thang Nguyen**, Soon-Yong Kwon, and Sung Youb Kim. “Auxeticity in Metals and Periodic Metallic Porous Structures Induced by Elastic Instabilities”. In: *physica status solidi (b)* 256.1 (2019), p. 1800122. DOI: 10.1002/pssb.201800122.

[11] **Duc Tam Ho**, **Cao Thang Nguyen**, Soon-Yong Kwon, and Sung Youb Kim. “Auxeticity in Metals and Periodic Metallic Porous Structures Induced by Elastic Instabilities (Phys. Status Solidi B 1/2019)”. In: *physica status solidi (b)* 256.1 (2019), p. 1970010. DOI: 10.1002/pssb.201970010.

*(The authors in **boldface** have equal contributions.)*

Manuscripts

[12] Cao Thang Nguyen and Sung Youb Kim. “ALFF: Automated Frameworks for Active-Learning Graph-Based Machine Learning Force Fields and Materials Property Calculations"”. 2025.

[13] Cao Thang Nguyen and Sung Youb Kim. “GraphNN-Based Interatomic Potential Enhancing Interlayer Interactions in Transition Metal Dichalcogenides”. 2025.

[14] Cao Thang Nguyen and Sung Youb Kim. “Mechanical and thermodynamic melting of metals from the mean-force dynamics calculation”. 2023.

My packages

[15] Cao Thang Nguyen. *"ALFF: Automated Frameworks for Active-Learning Graph-Based Machine Learning Force Fields and Materials Property Calculations"*. 2025.

[16] Cao Thang Nguyen. *"thml: Python package for LLM applications"*. 2023.

[17] Cao Thang Nguyen. *"thkit: Python package for general utilities."*. 2024.

[18] Cao Thang Nguyen. *"asext: Python package extends functions of ASE (Atomic Simulation Environment)"*. 2025.

[19] Cao Thang Nguyen. *"thmd: Python package for building models, pre-processing and post-processing Molecular Dynamics simulations."*. 2021.

[20] Cao Thang Nguyen. *"molbuilder: Online app to generate models for atomistic simulations"*. 2023.

[21] Cao Thang Nguyen. *"thang_foss: Fedora Copr repository to keep up-to-date builds of various cutting-edge packages for the latest RedHat based distros"*. 2025.

FOSS contributions

[22] Park et al. *SevenNet: Scalable Parallel Algorithm for Graph Neural Network Interatomic Potentials in Molecular Dynamics Simulations*. 11. (My contribution: PR#89, PR#92). 2024, pp. 4857–4868.

[23] Zhang et al. *DP-GEN: A concurrent learning platform for the generation of reliable deep learning based potential energy models*. (My contribution: PR#1545, PR#1556, PR#1563, PR#1627). 2020, p. 107206.

[24] Zhang et al. *DP-GEN2: the 2nd generation of the Deep Potential GENerator.* (My contribution: PR#324). 2025.

[25] Zeng et al. *Dpdata: A Scalable Python Toolkit for Atomistic Machine Learning Data Sets*. (My contribution: PR#614, PR#633). 2025.

[26] Yuan et al. *dpdispatcher is a Python package used to generate HPC (High Performance Computing) scheduler systems jobs input scripts and submit these scripts to HPC systems and poke until they finish*. (My contribution: PR#446, PR#470, PR#473, PR#475, PR#491, PR#501, PR#554). 2025.

[27] Larsen et al. *ASE: The atomic simulation environment—a Python library for working with atoms*. 27. (My contribution: PR#3529, PR#3623, PR#3847). June 2017, p. 273002.

[28] Klein et al. *"mBuild: A Hierarchical, Component Based Approach to Screening Properties of Soft Matter"*. (My contribution: PR#1098). Singapore, 2016, pp. 79–92.

REFEREES

i. Dr. Sung Youb Kim
   Professor, Department of Mechanical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, 44919, South Korea
   Phone: +82 52-217-2321
   Email: sykim@unist.ac.kr

ii. Dr. Duc Tam Ho
   Asst. Professor, Department of Mechanical and Construction Engineering, Northumbria University, Newcastle Upon Tyne, NE1 8ST, United Kingdom
   Phone: +44 7864-079-444
   Email: ductam.ho@northumbria.ac.uk